Introduction to Econometrics
Week 4

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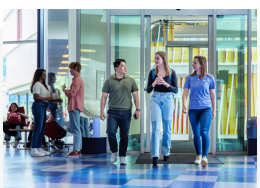
Based on lecture notes from previous instructors

February 23, 2026

university of groningen

faculty of economics and business

Master Orientation for 2nd year students



When: Monday, 2 March 2026, 15:15-16:30
Where: Duisenberg building, room 5412.0028

Programme:

15:15-16:30

- Presentations and pitches by current MSc students
- How to choose a master?
- What FEB MSc programmes can you study?
  And what are the entry requirements?
- What are the job prospects after graduation?

  ... and more!

  16:15 – 16:30
- Q&A with FEB MSc students

Sign up: www.rug.nl/feb/register

More information?
Contact us at prospectives.feb@rug.nl

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Main learning outcomes

After studying this lecture, you should be able to:

- derive the properties of the asymptotic properties of the OLS estimator (consistency, asymptotic normality etc.)
- demonstrate that (under some rather mild conditions) the OLS estimator $\mathbf{b}$ is always consistent for the least squares projection coefficients $\boldsymbol{\beta}^*$
- explain why endogeneity (i.e. failure of the predetermined regressors' assumption) only makes sense if you have first defined the coefficient of interest $\boldsymbol{\beta}$
- explain the difference between conditional homoskedasticity vs conditional heteroskedasticity
- to derive hypotheses tests for single coefficients, linear combinations of parameters and conditional homoskedasticity

Overview

1 Small vs large sample theory . . . 5

2 Consistency of OLS estimator . . . 10

2.1 Assumptions . . . 10

2.2 Derivation . . . 18

3 Projections . . . 23

4 Asymptotic distribution of OLS estimator . . . 30

4.1 Derivation . . . 30

4.2 Estimation of the asymptotic variance . . . 35

5 Hypotheses tests . . . 39

5.1 Single coefficient . . . 39

5.2 Linear combination of parameters . . . 42

5.3 Homoskedasticity . . . 46

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1 Small vs large sample theory . . . 5

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Assumptions and Results of Chapter 1

- **A1.1 (linearity)** : $y_i = \mathbf{x}_i^{'}\boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \ldots, n$
- **A1.2 (strict exogeneity)**: $\mathbb{E}(\varepsilon_i|\mathbf{X}) = 0, \quad i = 1, \ldots, n$
- **A1.3 (no perfect multicollinearity)**: p(column rank of $\mathbf{X}$=K) = 1

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- **A1.4 (spherical error variance)**: $\mathbb{E}(\varepsilon^2|\mathbf{X}) = \sigma^2\mathbf{I}_n$

Adding A1.4 $\Rightarrow$

1.
2.
3.

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- **A1.4 (spherical error variance)**: $\mathbb{E}(\varepsilon^2|\mathbf{X}) = \sigma^2\mathbf{I}_n$
- **A1.5 (normality of error term)**: $\varepsilon|\mathbf{X}$ is jointly normal

Adding A1.5 $\Rightarrow$

1.
2.

Advantages of large sample theory

It will enable us to weaken the following assumptions:

- strict exogeneity (especially relevant for time series)
- normality assumption
  - normality distribution of the OLS estimator is going to remain without assuming normally-distributed error terms (CLT-magic!)
  - so we do not need this assumption to do inference

We will also relax the spherical error variance assumption

- we can allow for conditional heteroskedasticity, which will lead to more general variance formulas

When will these results apply?

When a sample is large enough so that large sample theory provides a good approximation.

What is large enough? No real straight general answer here... Can be assessed using simulation methods for instance.

Depends also on the estimator...

Consistency vs unbiasedness

- $\mathbf{b}$ is unbiased for $\boldsymbol{\beta}$ if $\mathbb{E}\left[\mathbf{b}|\mathbf{X}\right] = \boldsymbol{\beta} \Rightarrow \mathbb{E}\left[\mathbf{b}\right] = \boldsymbol{\beta}$
- What does this intuitively mean? If we take many independent random samples of size n from the population, compute $\mathbf{b}$ for each, and do an average of these estimates, we will get very close to $\boldsymbol{\beta}$.
- So, in practice, if we average estimates from many different studies drawing small samples from the same population, we can get a reliable measure of the true $\boldsymbol{\beta}$.

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- So, in practice, if we average estimates from many different studies drawing small samples from the same population, we can get a reliable measure of the true $\boldsymbol{\beta}$.
- $\mathbf{b}$ is consistent for $\boldsymbol{\beta}$ if: $\underset{n\to\infty}{plim}\, \mathbf{b} = \boldsymbol{\beta}$
- Intuitively means that just one study with a large enough sample can give a reliable estimate of $\boldsymbol{\beta}$
- Important: a consistent estimator may be biased in small samples.

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The model and assumptions

- Suppose that we seek to estimate a well defined vector of coefficients $\boldsymbol{\beta}$ (or a subset of those) in a linear model of the form:

$$y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i, \; i = 1, \ldots, n$$

- Under some conditions are met, we will show that the OLS estimator is consistent for $\boldsymbol{\beta}$

- The key condition is that regressors are predetermined which means that:

$$\mathbb{E}\left[\mathbf{x}_i \varepsilon_i\right] = \mathbf{0}.$$

  If e.g. we are interested in measuring the causal effect of some variable x_{ik} on y_i,[1] then it will usually only hold if $\mathbf{x}_i$ includes sufficient variables that are correlated with x_{ik} and have a direct impact on y_i. Whether this is possible depends on the context. Otherwise, we need to rely on different estimation strategies (see next lectures).

- Importantly, violations of this condition only make sense if you define $\boldsymbol{\beta}$ first (see next section)

[1] Which means that we have defined our β_k of interest as measuring the causal effect of x_{ik} on y_i.

The model and assumptions

- **Data generating process (DGP)**: stochastic process that generated the finite sample $(\mathbf{y}, \mathbf{X})$
- If we specify the DGP, the joint distribution of the finite sample $(\mathbf{y}, \mathbf{X})$ can be determined.
- The model we study is the set of DGPs satisfying the set of assumptions that will follow

The model and assumptions

Assume that the following assumptions hold:

- **A2.1 (linearity):** $y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i,\ i = 1, \ldots, n$
- **A2.2' (i.i.d.):** $\{y_i, \mathbf{x}_i\}$ is an i.i.d. random sample
- **A2.3 (predetermined regressors):** Regressors are predetermined:

$$\mathbb{E}[\underbrace{\mathbf{x}_i\varepsilon_i}_{\equiv \mathbf{g}_i}] = \mathbb{E}[\mathbf{x}_i(y_i - \mathbf{x}_i'\boldsymbol{\beta})] = \mathbf{0}$$

- **A2.4 (rank condition):**

$$\boldsymbol{\Sigma}_{xx} \equiv \underset{(K\times K)}{\mathbb{E}(\mathbf{x}_i\mathbf{x}_i')}$$

  is nonsingular (and hence finite).

We will show that, if these assumptions hold

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i y_i\right) \underset{p}{\to} \boldsymbol{\beta}$$

Comments on the assumptions

1. A2.1. Typically, we can always write y_i as

$$y_i = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i.$$

To do that, one just needs to define ε_i as:

$$\varepsilon_i \equiv y_i - \mathbf{x}_i'\boldsymbol{\beta}$$

The key element to evaluate (and which is most often likely not met) is whether the error term constructed in this way satisfies some key properties for the OLS estimator to be consistent for $\boldsymbol{\beta}$.

Comments on the assumptions

2. A2.2' allows errors to be conditionally heteroskedastic. To see this:

$$
\begin{aligned}
&\{y_i, \mathbf{x}_i\} \text{ i.i.d.} \\
\Rightarrow &\{\varepsilon_i = y_i - \mathbf{x}_i'\boldsymbol{\beta}\} \text{ i.i.d.} \\
\Rightarrow &\mathbb{E}(\varepsilon_i^2) = \sigma^2, \ \forall i \qquad\qquad \text{(unconditional homoskedasticity)}
\end{aligned}
$$

This does not prevent $\mathbb{E}(\varepsilon_i^2|\mathbf{x}_i) = f(\mathbf{x}_i)$ where f is common to all i, indeed in this case:

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This does not prevent $\mathbb{E}(\varepsilon_i^2|\mathbf{x}_i) = f(\mathbf{x}_i)$ where f is common to all i, indeed in this case:

3. A2.2' still requires errors to be uncorrelated. Why?

Comments on the assumptions

4. (i) $\mathbb{E}(\mathbf{x}_i \varepsilon_i) = \mathbf{0}$ (orthogonality) vs. (ii) $\mathbb{E}(\varepsilon_i | \mathbf{x}_i) = 0$ (conditional mean independence)?

- (ii) is stronger than (i). Why?

Comments on the assumptions

4. (i) $\mathbb{E}(\mathbf{x}_i \varepsilon_i) = \mathbf{0}$ (orthogonality) vs. (ii) $\mathbb{E}(\varepsilon_i | \mathbf{x}_i) = 0$ (conditional mean independence)?

- (ii) is stronger than (i). Why?

- under i.i.d. assumption, how does $\mathbb{E}(\varepsilon_i | \mathbf{x}_i) = 0$ compare to $\mathbb{E}(\varepsilon_i | \mathbf{X}) = 0$?

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- under i.i.d. assumption, how does $\mathbb{E}(\varepsilon_i | \mathbf{x}_i) = 0$ compare to $\mathbb{E}(\varepsilon_i | \mathbf{X}) = 0$?

- Is it the same for time series?

Comments on the assumptions

5. If regressors include a constant, A2.3 ($\mathbb{E}(\mathbf{x}_i \varepsilon_i) = \mathbf{0}$) implies:

1. $\mathbb{E}(\varepsilon_i) = 0$
2. $Cov(\mathbf{x}_i, \varepsilon_i) = \mathbf{0}$ (regressors uncorrelated with error term)

Therefore, when the model features a constant, the predetermined regressors assumption is equivalent to assuming no correlation between the regressors and the error term.

Overview

1 Small vs large sample theory . . . 5

2 Consistency of OLS estimator . . . 10

2.1 Assumptions . . . 10

2.2 Derivation . . . 18

3 Projections . . . 23

4 Asymptotic distribution of OLS estimator . . . 30

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5.1 Single coefficient . . . 39

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Consistency of OLS estimator

Proposition 2.1.a. (Consistency of $\mathbf{b}$ for β):

Under A2.1-2.4:

$$\underset{n\to\infty}{plim}\,\mathbf{b} = \boldsymbol{\beta}$$

Proof steps

How to prove that?

1. start from OLS formula: $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i y_i\right)$
2. using A2.1. show that (left as an exercise):

$$\mathbf{b} - \boldsymbol{\beta} = \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right)$$

3. show: $1st$ term has a plim & $2nd$ has plim equal to $\mathbf{0}$, using ??
4. use Lemma 2.3 to show that RHS has plim equal to $\mathbf{0}$

Proof

$$\mathbf{b} - \boldsymbol{\beta} = \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \varepsilon_i \right)$$

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$$\mathbf{b} - \boldsymbol{\beta} = \underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i'\right)^{-1}}_{\xrightarrow[p]{}(\boldsymbol{\Sigma}_{xx})^{-1}}\underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right)}_{\xrightarrow[p]{}\mathbf{0}}$$

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1 Small vs large sample theory . . . 5

2 Consistency of OLS estimator . . . 10

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2.2 Derivation . . . 18

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4 Asymptotic distribution of OLS estimator . . . 30

4.1 Derivation . . . 30

4.2 Estimation of the asymptotic variance . . . 35

5 Hypotheses tests . . . 39

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Reminder of the assumptions

We showed that under the following assumptions, the OLS estimator $\mathbf{b}$ of $\boldsymbol{\beta}$ was consistent

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- **A2.4 (rank condition):** $\boldsymbol{\Sigma}_{xx} \equiv \mathbb{E}(\mathbf{x}_i\mathbf{x}_i')$ is nonsingular (and hence finite).

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We will now show that **A2.3** is necessarily true if the other assumptions hold and if you do not define $\boldsymbol{\beta}$ first. Another way to say that is that (under the other assumptions), the OLS estimator will always be consistent for something. This something is the least squares projection coefficient.

Predictions

- Assume that **A2.2' (i.i.d)** and **A2.4 (rank condition)** hold
- Consider the goal of **predicting** y_i from $\mathbf{x}_i$ based on sample $\{y_i, \mathbf{x}_i\}$
- A **predictor** is a function $g(\mathbf{x})$ where $g(.)$ determined by the joint distribution of $(y, \mathbf{x})$
- **Forecast error**: $e = y - g(\mathbf{x})$
- **Loss associated with forecast error**: $L(e)$ increasing in $|e|$
- **Expected loss**: $\mathbb{E}[L(e)]$
- The aim of prediction is to minimize expected loss
- Optimal prediction depends on L and on the set of functions g considered

Optimal *linear* predictor with squared error loss

- Consider prediction with a squared error loss function $L(e) = e^2$ and in the case where we restrict g to be linear in $\mathbf{x}$
- That is, we restrict ourselves to functions of the form:

$$g(\mathbf{x}) = \mathbf{x}'\tilde{\boldsymbol{\beta}}$$

  and we look for:

$$\boldsymbol{\beta}^* = \arg\min_{\tilde{\boldsymbol{\beta}}} \mathbb{E}\left[\left(y - \mathbf{x}'\tilde{\boldsymbol{\beta}}\right)^2\right]$$

- The solution to this problem is the least squares projection coefficient:[2]

$$\boldsymbol{\beta}^* = \left[\mathbb{E}\left(\mathbf{x}\mathbf{x}'\right)\right]^{-1}\mathbb{E}\left(\mathbf{x}y\right)$$

[2] Note that A2.4 is needed here.

Properties of the prediction error

- The prediction error from least squares projection is:

$$e = y - \mathbf{x}'\boldsymbol{\beta}^*$$

- using the expression for $\boldsymbol{\beta}^* = \left[\mathbb{E}\left(\mathbf{x}\mathbf{x}'\right)\right]^{-1}\mathbb{E}\left(\mathbf{x}y\right)$, we have:

$$\mathbb{E}(\mathbf{x}e) = \mathbb{E}(\mathbf{x}y) - \mathbb{E}(\mathbf{x}\mathbf{x}')\boldsymbol{\beta}^* = \mathbb{E}(\mathbf{x}y) - \mathbb{E}(\mathbf{x}y) = \mathbf{0}$$

- Therefore, we can write y as:

$$y = \mathbf{x}'\boldsymbol{\beta}^* + e$$

  with

$$\mathbb{E}(\mathbf{x}e) = \mathbb{E}\left(\mathbf{x}\left(y - \mathbf{x}'\boldsymbol{\beta}^*\right)\right) = \mathbf{0}$$

  where $\boldsymbol{\beta}^*$ is the LS projection coefficient and e is the prediction error from the LS projection. What do you notice?

Least squares projection coefficients in terms of variances and covariances

If one of regressors is a constant ($\mathbf{x} = (1, \widetilde{\mathbf{x}}')'$), we can write least squares projection coefficients in terms of variances and covariances[3]

$$\mathbb{P}(y|\mathbf{x}) \equiv \mathbb{P}(y|1, \widetilde{\mathbf{x}}) = \mu + \boldsymbol{\gamma}'\widetilde{\mathbf{x}}$$

where $\boldsymbol{\gamma} = \text{Var}(\widetilde{\mathbf{x}})^{-1}\,\text{Cov}(\widetilde{\mathbf{x}}, y)$ and $\mu = \mathbb{E}(y) - \boldsymbol{\gamma}'\mathbb{E}(\widetilde{\mathbf{x}})$ proof

As OLS consistently estimate LS projection coefficients, we can interpret OLS as capturing (asymptotically) all the correlation between $\mathbf{x}$ and y.

Another way to see that is that projection with intercept are defined by:

$$cov(\mathbf{x}, e) = \mathbf{0}$$

So that there is nothing left in the residual that is correlated with $\mathbf{x}$.

[3] Hayashi uses the notation $\widehat{\mathbb{E}}^*$ instead of $\mathbb{P}$ for the linear projection. This is because the expectation is the best predictor with MSE loss (see proposition 2.7 p138), and because the projection is the best predictor of the expectation with MSE loss.

Example

- to estimate the causal effect of education on earnings, we may want to estimate by OLS a model of the form:

$$earnings_i = \beta_0 + \beta_1 educ_i + \varepsilon_i$$

- we, therefore, would like the LS projection coefficient of earnings on education and β_1 (measuring the causal effect of education on earnings) to coincide

- one issue is that ability may have a direct impact on earnings and is likely correlated with education, so that a model of the form:

$$earnings_i = \tilde{\beta}_0 + \beta_1 educ_i + \beta_2 abil_i + u_i$$

  is plausible (where β_2 measures the causal impact of ability on earnings).[4] Assuming that $\mathbb{E}\left(u_i \begin{pmatrix} educ_i \\ abil_i \end{pmatrix}\right) = \mathbf{0}$, we have:

  - $\varepsilon_i =$ $\Rightarrow$

[4] As we are not interested in the constant we define $\beta_0 = \mathbb{E}(earnings) - \beta_1 \mathbb{E}(educ)$ and $\tilde{\beta}_0 = \mathbb{E}(earnings) - \beta_1 \mathbb{E}(educ) - \beta_2 \mathbb{E}(abil)$, so that the error terms ε_i and u_i both have an expectation of 0.

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2 Consistency of OLS estimator . . . 10

2.1 Assumptions . . . 10

2.2 Derivation . . . 18

3 Projections . . . 23

4 Asymptotic distribution of OLS estimator . . . 30

4.1 Derivation . . . 30

4.2 Estimation of the asymptotic variance . . . 35

5 Hypotheses tests . . . 39

5.1 Single coefficient . . . 39

5.2 Linear combination of parameters . . . 42

5.3 Homoskedasticity . . . 46

Asymptotic distribution of OLS estimator

- under **A2.1**-**A2.4**, we have seen that the OLS estimator converges in probability to $\boldsymbol{\beta}$
- in practice, we work with rather large but far from infinite samples (think of survey data)
- therefore, important to have a measure of the uncertainty around our point estimate in order to do statistical inference
- that is what we seek to do now (assuming that our sample is large enough for large sample theory to provide a good approximation)
- to do that, which theorem do you think we will use? What extra assumption will be needed?

Proof steps

How are we going to do that?

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1. We are still going to consider the formula:

$$\mathbf{b} - \boldsymbol{\beta} = \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \varepsilon_i \right)$$

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2. We know that $E(\mathbf{x}_i \varepsilon_i) = \mathbf{0}$, so we can apply Lindberg-Levy CLT to the 2nd term assuming $Var(\mathbf{x}_i \varepsilon_i)$ exists and is finite

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3. Use lemma 2.4 to find the asymptotic distribution of the OLS estimator, given that we know the plim of the 1st term

Proof

$$\mathbf{b} - \boldsymbol{\beta} = \underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i'\right)^{-1}}_{\xrightarrow[p]{}(\boldsymbol{\Sigma}_{xx})^{-1}} \left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right)$$

- **A2.5'**: the following exists and is finite

$$\mathbf{S} = \text{Var}\left(\mathbf{x}_i\varepsilon_i\right) = E\left(\mathbf{x}_i\varepsilon_i\left(\mathbf{x}_i\varepsilon_i\right)'\right) = E\left(\varepsilon_i^2\mathbf{x}_i\mathbf{x}_i'\right)$$

Proof

$$\sqrt{n}\left(\mathbf{b}-\boldsymbol{\beta}\right)=\underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i'\right)^{-1}}_{\xrightarrow[p]{}(\boldsymbol{\Sigma}_{xx})^{-1}}\underbrace{\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right)}_{\xrightarrow[d]{}\mathcal{N}(\mathbf{0},\mathbf{S})}$$

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Proof

$$\sqrt{n}\left(\mathbf{b}-\boldsymbol{\beta}\right)=\underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i'\right)^{-1}}_{\xrightarrow[p]{}(\boldsymbol{\Sigma}_{xx})^{-1}}\underbrace{\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i\right)}_{\xrightarrow[d]{}\mathcal{N}(\mathbf{0},\mathbf{S})}\xrightarrow[d]{}\mathcal{N}\left(\mathbf{0},\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{S}\boldsymbol{\Sigma}_{xx}^{-1}\right)$$

- **A2.5'**: the following exists and is finite

$$\mathbf{S}=\text{Var}\left(\mathbf{x}_i\varepsilon_i\right)=E\left(\mathbf{x}_i\varepsilon_i\left(\mathbf{x}_i\varepsilon_i\right)'\right)=E\left(\varepsilon_i^2\mathbf{x}_i\mathbf{x}_i'\right)$$

Asymptotic Distribution of OLS Estimator

We have proved:

Proposition 2.1.b (Asymptotic Normality of *b*):

Under A2.1 to A2.5', the OLS estimator $\mathbf{b}$ is asymptotically normal with asymptotic variance[5]

$$Avar(\mathbf{b}) = \mathbf{\Sigma}_{xx}^{-1}\mathbf{S}\mathbf{\Sigma}_{xx}^{-1}$$

In other words

$$\sqrt{n}\,(\mathbf{b} - \boldsymbol{\beta}) \underset{d}{\rightarrow} N(\mathbf{0}, \mathbf{\Sigma}_{xx}^{-1}\mathbf{S}\mathbf{\Sigma}_{xx}^{-1})$$

Or

$$\mathbf{b} \underset{a}{\sim} \mathcal{N}\left(\boldsymbol{\beta}, \frac{1}{n}\mathbf{\Sigma}_{xx}^{-1}\mathbf{S}\mathbf{\Sigma}_{xx}^{-1}\right)$$

[5] Note that some authors call the asymptotic variance of $\mathbf{b}$ the variance of $\sqrt{n}\,(\mathbf{b} - \boldsymbol{\beta})$, i.e. $\mathbf{\Sigma}_{xx}^{-1}\mathbf{S}\mathbf{\Sigma}_{xx}^{-1}$ here. While others denote by this term the variance of $(\mathbf{b} - \boldsymbol{\beta})$, i.e. $n^{-1}\mathbf{\Sigma}_{xx}^{-1}\mathbf{S}\mathbf{\Sigma}_{xx}^{-1}$ here. Here, we follow Hayashi who uses the first definition (although my personal preference is for the second one).

Overview

1 Small vs large sample theory . . . 5

2 Consistency of OLS estimator . . . 10

2.1 Assumptions . . . 10

2.2 Derivation . . . 18

3 Projections . . . 23

4 Asymptotic distribution of OLS estimator . . . 30

4.1 Derivation . . . 30

4.2 Estimation of the asymptotic variance . . . 35

5 Hypotheses tests . . . 39

5.1 Single coefficient . . . 39

5.2 Linear combination of parameters . . . 42

5.3 Homoskedasticity . . . 46

Estimation of the asymptotic variance

- the true asymptotic variance:
$$Avar(\mathbf{b}) = \mathbf{\Sigma}_{xx}^{-1}\mathbf{S}\mathbf{\Sigma}_{xx}^{-1}$$
is unobserved, so we need to find a consistent estimator for it

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- so if we have available $\hat{\mathbf{S}}$ such that: $\hat{\mathbf{S}} \underset{p}{\longrightarrow} \mathbf{S}$

- then by lemma 2.3:
$$\widehat{Avar}(\mathbf{b}) = \mathbf{S}_{xx}^{-1}\hat{\mathbf{S}}\mathbf{S}_{xx}^{-1} \underset{p}{\longrightarrow} Avar(\mathbf{b}) = \mathbf{\Sigma}_{xx}^{-1}\mathbf{S}\mathbf{\Sigma}_{xx}^{-1}$$
which is **proposition 2.1c**.

- $\mathbf{S} = E\left(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i'\right)$. If we observed ε_i, a consistent estimate for $\mathbf{S}$ would be easy as $\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i'$ is i.i.d. and $\mathbf{S}$ finite by A2.5'.

[6] See section 2.5 in Hayashi for further details.

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- In that case, Kolmogorov's LLN would imply:

$$\hat{\mathbf{S}} = \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i' \underset{p}{\longrightarrow} \mathbf{S} = E\left(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i'\right)$$

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- under an additional (fourth-moment) assumption:

$$\textbf{A2.6}\text{: } E\left[\left(x_{ij} x_{ik}\right)^2\right] \text{ is finite for all } k,\ j$$

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- the natural empirical equivalent which replaces ε_i by e_i: [6]

$$\hat{\mathbf{S}} = \frac{1}{n} \sum_{i=1}^{n} e_i^2 \mathbf{x}_i \mathbf{x}_i' \underset{p}{\longrightarrow} \mathbf{S} = E\left(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i'\right)$$

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- one can also use a variant, with possibly better finite sample properties, which divides by $n - K$ instead of n.

[6] See section 2.5 in Hayashi for further details.

Estimation of the asymptotic variance

- if we impose the (strong) assumption of (condtional) homoskedasticity we do not need A2.5' and A2.6.
- left as an exercise to show this and that in this case:

$$Avar\left(\mathbf{b}\right) = \mathbf{\Sigma}_{xx}^{-1}\mathbf{S}\mathbf{\Sigma}_{xx}^{-1} = \sigma^2\mathbf{\Sigma}_{xx}^{-1}$$

- in that case $\widehat{Avar\left(\mathbf{b}\right)} = s^2\mathbf{S}_{xx}^{-1}$ is a consistent estimator of $Avar\left(\mathbf{b}\right)$
- indeed (**proposition 2.2**):

$$s^2 = \frac{1}{n-K}\sum_{i=1}^{n} e_i^2 \underset{p}{\longrightarrow} E\left(\varepsilon_i^2\right)$$

  under A2.1 to A2.4 and provided $E\left(\varepsilon_i^2\right)$ exists and is finite.
- true for every $e_i \equiv y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}$ provided $\hat{\boldsymbol{\beta}}$ consistent for $\boldsymbol{\beta}$

Overview

1 Small vs large sample theory . . . 5

2 Consistency of OLS estimator . . . 10

2.1 Assumptions . . . 10

2.2 Derivation . . . 18

3 Projections . . . 23

4 Asymptotic distribution of OLS estimator . . . 30

4.1 Derivation . . . 30

4.2 Estimation of the asymptotic variance . . . 35

5 Hypotheses tests . . . 39

5.1 Single coefficient . . . 39

5.2 Linear combination of parameters . . . 42

5.3 Homoskedasticity . . . 46

Hypothesis test for a single coefficient

- from previously, under H0: $\beta_k = \overline{\beta}_k$:

$$\sqrt{n}\left(b_k - \overline{\beta}_k\right) \underset{d}{\longrightarrow} \mathcal{N}\left(0, Avar\left(b_k\right)\right)$$

$$\widehat{Avar\left(b_k\right)} = \left(\mathbf{S}_{xx}^{-1}\hat{\mathbf{S}}\mathbf{S}_{xx}^{-1}\right)_{kk} \underset{p}{\longrightarrow} Avar\left(b_k\right)$$

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- using lemma 2.4c, this implies (**proposition 2.3a**):

$$t_k \equiv \frac{b_k - \overline{\beta}_k}{SE\left(b_k\right)} \underset{d}{\longrightarrow} \mathcal{N}\left(0, 1\right)$$

with

$$SE\left(b_k\right) \equiv \sqrt{\frac{1}{n}\widehat{Avar\left(b_k\right)}} = \sqrt{\frac{1}{n}\left(\mathbf{S}_{xx}^{-1}\hat{\mathbf{S}}\mathbf{S}_{xx}^{-1}\right)_{kk}}$$

heteroskedasticity-consistent, (heteroskedasticity-)robust or White's standard error

Hypothesis test for a single coefficient

Hence to test H0:

1. calculate t_k
2. look at the critical value $t_{\alpha/2}$ for $\mathcal{N}(0,1)$
3. reject H0 if $|t_k| > t_{\alpha/2}$

Hypothesis test for a single coefficient

Hence to test H0:

1. calculate t_k
2. look at the critical value $t_{\alpha/2}$ for $\mathcal{N}(0,1)$
3. reject H0 if $|t_k| > t_{\alpha/2}$

- actual size of test = desired size (α) only approximately
- approximation very good as sample size very large
- size distortion = actual size - desired size

Overview

1 Small vs large sample theory . . . 5

2 Consistency of OLS estimator . . . 10

2.1 Assumptions . . . 10

2.2 Derivation . . . 18

3 Projections . . . 23

4 Asymptotic distribution of OLS estimator . . . 30

4.1 Derivation . . . 30

4.2 Estimation of the asymptotic variance . . . 35

5 Hypotheses tests . . . 39

5.1 Single coefficient . . . 39

5.2 Linear combination of parameters . . . 42

5.3 Homoskedasticity . . . 46

Hypothesis test for linear combinations of parameters

- we are sometimes interested in testing hypotheses of the form:

$$\mathbf{R}\boldsymbol{\beta} = \mathbf{r} \Leftrightarrow \mathbf{R}\boldsymbol{\beta} - \mathbf{r} = \mathbf{0}$$

  where $\mathbf{R}$ is $\#r \times K$ and $\mathbf{r}$ is $\#r \times 1$.
- to do so, need to convert these multiple restrictions into a single statistic with known distribution

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- to do so, need to convert these multiple restrictions into a single statistic with known distribution

- typically, as we have a normally distributed vector (as OLS estimator is normally distributed), we are going to take a quadratic transformation of the vector of restrictions and this is going to be distributed χ^2

1. start from the distribution of OLS estimator and multiply by $\mathbf{R}$:

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3. furthermore:

$$\mathbf{R}\widehat{Avar}\left(\mathbf{b}\right)\mathbf{R}' \underset{p}{\longrightarrow} \mathbf{R}Avar\left(\mathbf{b}\right)\mathbf{R}'$$

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3. furthermore:
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4. form the quadratic form (which requires $\mathbf{R}$ to be of full row rank) and use lemma 2.4d. (reminder: $\underset{(n\times 1)}{\mathbf{x}} \sim \mathcal{N}\left(\mathbf{0}, \mathbf{\Sigma}\right) \Rightarrow \mathbf{x}'\mathbf{\Sigma}^{-1}\mathbf{x} \sim \chi^2\left(n\right)$):

Link with test statistics from previous classes

- if we assume conditional homoskedasticity as in previous class, we can show that:

$$\frac{b_k - \beta_k}{\sqrt{s^2 \left(\mathbf{X}'\mathbf{X}\right)^{-1}_{kk}}} \xrightarrow[d]{} \mathcal{N}(0,1)$$

- so under conditional homoskedasticity the t-ratio from current and previous classes both converge to the same asymptotic distribution

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- so under conditional homoskedasticity the t-ratio from current and previous classes both converge to the same asymptotic distribution

- furthermore, under this same assumption:

$$\#\mathbf{r} \times F \underset{d}{\longrightarrow} \chi^2 \left(\#\mathbf{r}\right)$$

  where F is the F-statistic from previous classes

- so the Wald and (slightly adjusted) F statistics have the same asymptotic distribution

Overview

1 Small vs large sample theory . . . 5

2 Consistency of OLS estimator . . . 10

2.1 Assumptions . . . 10

2.2 Derivation . . . 18

3 Projections . . . 23

4 Asymptotic distribution of OLS estimator . . . 30

4.1 Derivation . . . 30

4.2 Estimation of the asymptotic variance . . . 35

5 Hypotheses tests . . . 39

5.1 Single coefficient . . . 39

5.2 Linear combination of parameters . . . 42

5.3 Homoskedasticity . . . 46

Testing for conditional homoskedasticity

- under A2.1-A2.6:

$$\hat{\mathbf{S}} = \frac{1}{n}\sum_{i=1}^{n} e_i^2 \mathbf{x}_i \mathbf{x}_i' \underset{p}{\longrightarrow} \mathbf{S}$$

- the additional hypothesis of conditional homoskedsaticity:

$$s^2 \mathbf{S}_{xx} = \frac{1}{n}\sum_{i=1}^{n} s^2 \mathbf{x}_i \mathbf{x}_i' \underset{p}{\longrightarrow} \mathbf{S}$$

- therefore, under H0 of conditional homoskedsaticity:

$$\hat{\mathbf{S}} - s^2 \mathbf{S}_{xx} \underset{p}{\longrightarrow} \mathbf{0} \Rightarrow \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i' \left(e_i^2 - s^2\right) \underset{p}{\longrightarrow} \mathbf{0}$$

$$\Rightarrow \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{\psi}_i \left(e_i^2 - s^2\right) \underset{p}{\longrightarrow} \mathbf{0}$$

where $\boldsymbol{\psi}_i$ $(m \times 1)$ made of the unique and non-constant elements of $\mathbf{x}_i \mathbf{x}_i'$

Testing for conditional homoskedasticity

$$\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{\psi}_i \left(e_i^2 - s^2\right) \underset{p}{\longrightarrow} \mathbf{0}$$

- why not keeping constant elements?

Testing for conditional homoskedasticity

$$\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{\psi}_i\left(e_i^2 - s^2\right) \underset{p}{\longrightarrow} \mathbf{0}$$

- why not keeping constant elements?
  - because constant will be in regression below
  - also, condition for constant implies that:

$$\frac{1}{n}\sum_{i=1}^{n}e_i^2 - s^2 \underset{p}{\longrightarrow} 0$$

which is true by construction of s^2

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    which is true by construction of s^2

- why not keeping non-unique elements?
  - because they do not provide any more information

Testing for conditional homoskedasticity

Based on this, White derives the following test:

1. regress e_i^2 on a constant and on ψ_i
2. compute the R^2
3. under H0: $nR^2 \underset{d}{\longrightarrow} \chi^2(m)$
4. hence, reject H0 if the previous statistic is too large, in which case non robust t and F statistics should NOT be used.

Readings from Hayashi

- 2.9
- 2.3
- 2.4 except section on asymptotic power and tests of nonlinear hypotheses
- 2.5
- 2.6
- 2.7
- for recap on asymptotic theory: see notes + ch 2.1 except delta method

Proof

As $\begin{pmatrix} \mu & \gamma \end{pmatrix}'$ are projection coefficients, they are defined as solving the orthogonality conditions:

$$\mathbb{E}\left[\mathbf{x}\left(y - \mathbf{x}'\begin{pmatrix} \mu \\ \gamma \end{pmatrix}\right)\right] = \mathbf{0}$$

$$\Rightarrow \mathbb{E}\left[\begin{pmatrix} 1 \\ \tilde{\mathbf{x}} \end{pmatrix}\begin{pmatrix} 1 & \tilde{\mathbf{x}} \end{pmatrix}\right]\begin{bmatrix} \mu \\ \gamma \end{bmatrix} = \mathbb{E}\left[\begin{pmatrix} 1 \\ \tilde{\mathbf{x}} \end{pmatrix} y\right]$$

$$\Rightarrow \begin{bmatrix} 1 & \mathbb{E}\left(\tilde{\mathbf{x}}\right)' \\ \mathbb{E}\left(\tilde{\mathbf{x}}\right) & \mathbb{E}\left(\tilde{\mathbf{x}}\tilde{\mathbf{x}}'\right) \end{bmatrix}\begin{bmatrix} \mu \\ \gamma \end{bmatrix} = \begin{bmatrix} \mathbb{E}\left(y\right) \\ \mathbb{E}\left(\tilde{\mathbf{x}}y\right) \end{bmatrix}$$

Proof

$$\begin{bmatrix} 1 & \mathbb{E}\left(\tilde{\mathbf{x}}\right)' \\ \mathbb{E}\left(\tilde{\mathbf{x}}\right) & \mathbb{E}\left(\tilde{\mathbf{x}}\tilde{\mathbf{x}}'\right) \end{bmatrix} \begin{bmatrix} \mu \\ \boldsymbol{\gamma} \end{bmatrix} = \begin{bmatrix} \mathbb{E}\left(y\right) \\ \mathbb{E}\left(\tilde{\mathbf{x}}y\right) \end{bmatrix}$$

So

$$\mu = \mathbb{E}\left(y\right) - \mathbb{E}\left(\tilde{\mathbf{x}}\right)' \boldsymbol{\gamma}$$

and

$$\mathbb{E}\left(\tilde{\mathbf{x}}\right) \underbrace{\mu}_{\mathbb{E}(y)-\mathbb{E}(\tilde{\mathbf{x}})'\boldsymbol{\gamma}} + \mathbb{E}\left(\tilde{\mathbf{x}}\tilde{\mathbf{x}}'\right) \boldsymbol{\gamma} = \mathbb{E}\left(\tilde{\mathbf{x}}y\right)$$

$$\Rightarrow \underbrace{\left(\mathbb{E}\left(\tilde{\mathbf{x}}\tilde{\mathbf{x}}'\right) - \mathbb{E}\left(\tilde{\mathbf{x}}\right)\mathbb{E}\left(\tilde{\mathbf{x}}\right)'\right)}_{var(\tilde{\mathbf{x}})} \boldsymbol{\gamma} = \underbrace{\mathbb{E}\left(\tilde{\mathbf{x}}y\right) - \mathbb{E}\left(\tilde{\mathbf{x}}\right)\mathbb{E}\left(y\right)}_{cov(\tilde{\mathbf{x}},y)}$$

$$\Rightarrow \boldsymbol{\gamma} = var\left(\tilde{\mathbf{x}}\right)^{-1} cov\left(\tilde{\mathbf{x}}, y\right)$$

back